

The internal process is helpful in the DevTest environment as we can instrument automatically in the background during any testing process. The external process is useful in production environments, allowing the instrumentation to be switched on/off at any time to collect telemetry data. Also, external processes can run before application warm start runs and continue even after the EKS service stops. This is helpful to understand performance issues through the invocation point for the bootstrap function (request/response).

Lightrun is handy for extended observability, metrics analytics and performance troubleshooting. It helps to analyse the filtered metrics by quickly setting up dashboards for unified observability and monitoring including container, software life cycle and IoT monitoring services. It can be integrated with Amazon Simple Notification Service (SNS) through topic subscription so that any notification from metrics analysis can be channelled to SNS notification alerts.

Dynamic instrumentation for logging and monitoring in an EKS application

Kubernetes containers run as an integrated package of various stacks comprising infrastructure service, platform service and application service. Infrastructure service handles runtime OS layer services including container orchestration and infrastructure components like compute, storage and network services.

Platform services contain application runtime environments including Spring Boot for Java applications, bootstrap services for application runtime environment, security services and orchestration engines for containers. Both infrastructure services and platform services are enablers for applications to run in the container.

Application services include Spring Boot applications running on an EKS instance, as well as integrated

management and monitoring services for native observability.

Using Lightrun, we can instrument the application runtime binary dynamically without restarting the application or adding debug/log statements to build/containerise application code.

If there aren't enough logs added in the running Java container (e.g., Spring Boot application in EKS instance), we can just install the Lightrun agent and attach it to the runtime application by adding the command-line argument *agentpath:<path>/lightrun_agent.dll*, and run the application. This will enable bootstrapping observability agents to the application startup.

Lightrun can be integrated with IDEs like IntelliJ IDEA or VSCode. It can be used from on-premises deployable service or as Lightrun cloud service (PaaS model) to integrate with cloud native applications including Java, Python or Scala applications running in containers like EKS, AKS or GKE.

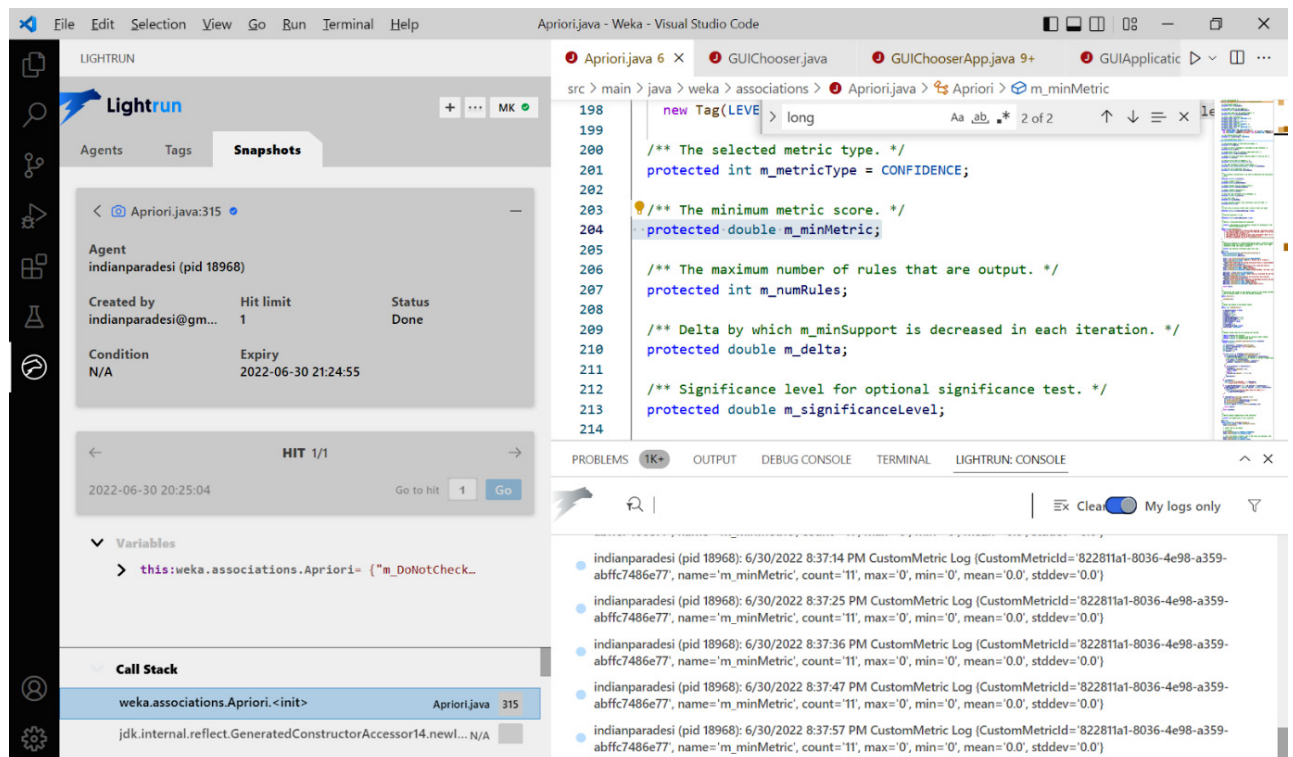


Figure 5: Snapshot of a running component in EKS using Lightrun agent